BIG DATA

Prof. Me. ***Orlando*** *da Silva Junior*

Big data *is extremely overhyped and not terribly well defined. Many people think they have big data, when they actually don’t.*



Hadley Wickham

Chief Scientist, Rstudio

Criador dos pacotes ggplot2 e Stats

I think there are **two particularly important transition points**:

1. **From in-memory to disk.** If your data fits in memory, it's small data. And these days you can get 1 TB of ram, so even small data is big! Moving from in-memory to on-disk is an important transition because access speeds are so different. You can do quite naive computations on in-memory data and it'll be fast enough. You need to plan (and index) much more with on-disk data;

2. **From one computer to many computers.** The next important threshold occurs when you data no longer fits on one disk on one computer. Moving to a distributed environment makes computation much more challenging because you don't have all the data needed for a computation in one place. Designing distributed algorithms is much harder, and you're fundamentally limited by the way the data is split up between computers.

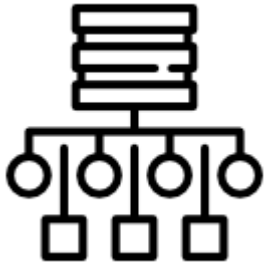
I personally believe it's impossible for one system to span from in-memory to on-disk to distributed. R is a fantastic environment for the rapid exploration of in-memory data, but there's no elegant way to scale it to much larger datasets. Hadoop works well when you have thousands of computers, but is incredible slow on just one machine. Fortunately, I don't think one system needs to solve all big data problems.

To me there are **three main classes of problem**:

1. **Big data problems that are actually small data problems**, once you have the right subset/sample/summary. Inventing numbers on the spot, I'd say 90% of big data problems fall into this category. To solve this problem you need a distributed database (like hive, impala, teradata etc), and a tool like dplyr to let you rapidly iterate to the right small dataset (which still might be gigabytes in size).

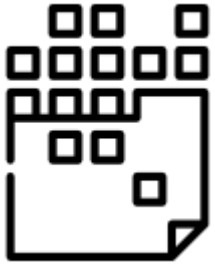
2. **Big data problems that are actually lots and lots of small data problems**, e.g. you need to fit one model per individual for thousands of individuals. I'd say ~9% of big data problems fall into this category. This sort of problem is known as a trivially parallelisable problem and you need some way to distribute computation over multiple machines. The foreach package is a nice solution to this problem because it abstracts away the backend, allowing you to focus on the computation, not the details of distributing it.

3. **Finally, there are irretrievably big problems** where you do need all the data, perhaps because you fitting a complex model. An example of this type of problem is recommender systems which really do benefit from lots of data because they need to recognise interactions that occur only rarely. These problems tend to be solved by dedicated systems specifically designed to solve a particular problem.



Volume

300 horas de vídeos são carregadas no YouTube por minuto.



Velocidade

Google processa mais de 3,5 bilhões de consultas de busca por dia.



Variedade

Mais de 80% dos dados do mundo são não-estruturados.



Veracidade



Variabilidade



Valor

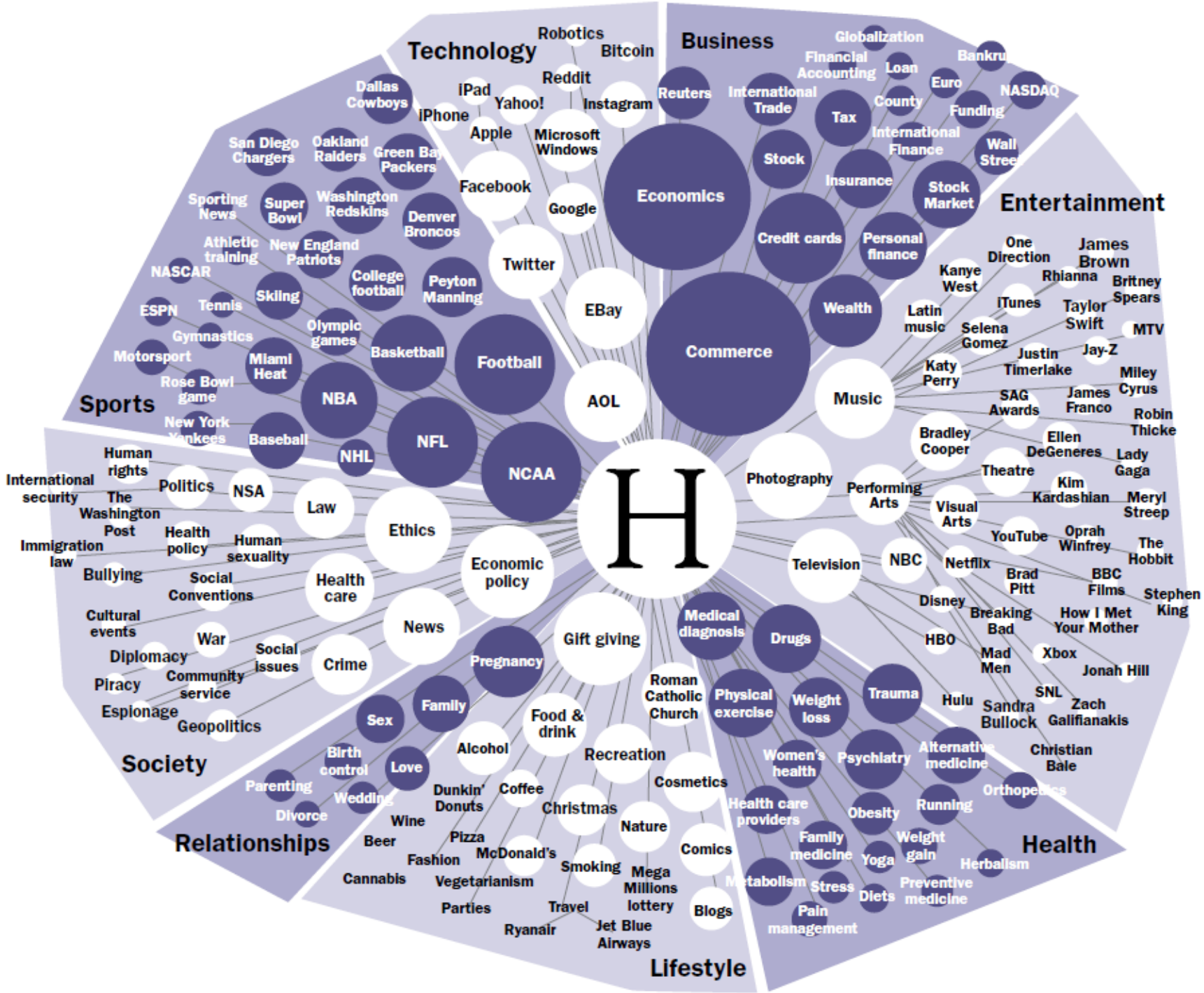


Visualização

THE HUFFINGTON POST

“HuffPo uses Big Data to optimise content, authenticate comments, ensure efficacy of native advertising, regulate advertising placement and create passive personalisation” (2014)
-- Jimmy Maymann, CEO

- ✓ 90 milhões de **usuários**
- ✓ 22 milhões de **visitas mensais**
- ✓ 110 milhões de visualizações de **vídeos** por mês
- ✓ 70 mil **bloggers**, incluindo Hillary Clinton, Deepak Chopra e Desmond Tutu


H
Technology
Robotics
Bitcoin
Reddit
iPad
Yahoo!
Instagram
iPhone
Apple
Microsoft Windows
Facebook
Google
Twitter
Business
Globalization
Financial Accounting
Loan
Bankru
Euro
NASDAQ
Reuters
International Trade
Tax
County
Funding
International Finance
Wall Stree
Stock
Insurance
Stock Market
Economics
Credit cards
Personal finance
Wealth
Commerce
EBay
AOL
Entertainment
One Direction
James Brown
Kanye West
Rhianna
Britney Spears
iTunes
Taylor Swift
Latin music
Selena Gomez
MTV
Justin Timerlake
Jay-Z
Katy Perry
Miley Cyrus
Music
SAG Awards
James Franco
Robin Thicke
Bradley Cooper
Ellen DeGeneres
Lady Gaga
Theatre
Kim Kardashian
Photography
Performing Arts
Visual Arts
Meryl Streep
YouTube
Oprah Winfrey
The Hobbit
Television
NBC
Netflix
Brad Pitt
BBC Films
Stephen King
Disney
Breaking Bad
How I Met Your Mother
HBO
Mad Men
Xbox
Jonah Hill
Hulu
SNL
Sandra Bullock
Zach Galifianakis
Christian Bale
Sports
Dallas Cowboys
San Diego Chargers
Oakland Raiders
Green Bay Packers
Sporting News
Super Bowl
Washington Redskins
Denver Broncos
Athletic training
New England Patriots
NASCAR
College football
Peyton Manning
ESPN
Tennis
Skiing
Gymnastics
Olympic games
Motorsport
Miami Heat
Basketball
Football
Rose Bowl game
NBA
NFL
New York Yankees
Baseball
NHL
NCAA
Society
Human rights
International security
Politics
NSA
The Washington Post
Law
Ethics
Immigration law
Health policy
Human sexuality
Bullying
Social Conventions
Health care
Economic policy
Cultural events
War
Diplomacy
Social issues
News
Crime
Piracy
Community service
Espionage
Geopolitics
Relationships
Sex
Family
Pregnancy
Gift giving
Birth control
Love
Parenting
Wedding
Divorce
Lifestyle
Alcohol
Food & drink
Recreation
Roman Catholic Church
Cosmetics
Dunkin' Donuts
Coffee
Christmas
Nature
Comics
Wine
Beer
Pizza
McDonald's
Fashion
Smoking
Mega Millions lottery
Cannabis
Vegetarianism
Parties
Travel
Ryanair
Jet Blue Airways
Blogs
Health
Medical diagnosis
Drugs
Physical exercise
Weight loss
Trauma
Women's health
Psychiatry
Alternative medicine
Orthopedics
Health care providers
Obesity
Running
Family medicine
Yoga
Weight gain
Herbalism
Metabolism
Stress
Diets
Preventive medicine
Pain management

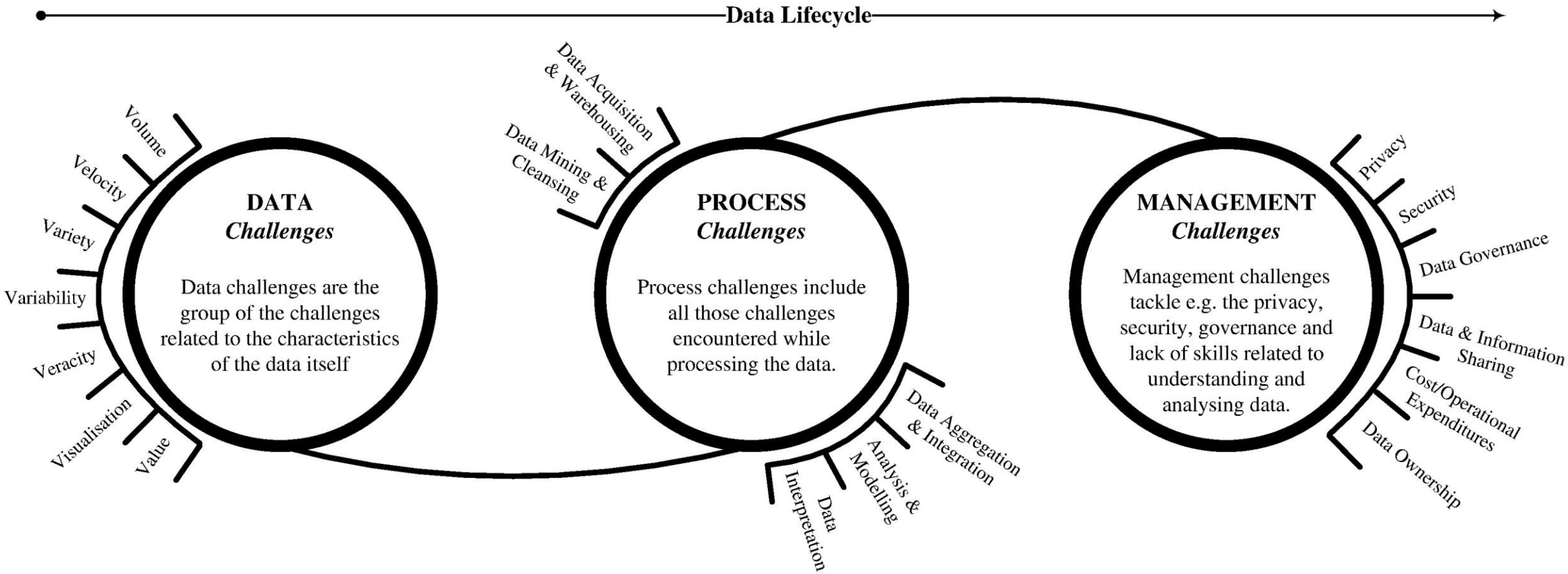
Ministério da Justiça

Para ajudar a combater a **lavagem de dinheiro**, as autoridades brasileiras do Ministério da Justiça buscavam identificar e investigar rendimentos ilícitos em uma velocidade maior do que a da atividade criminosa.

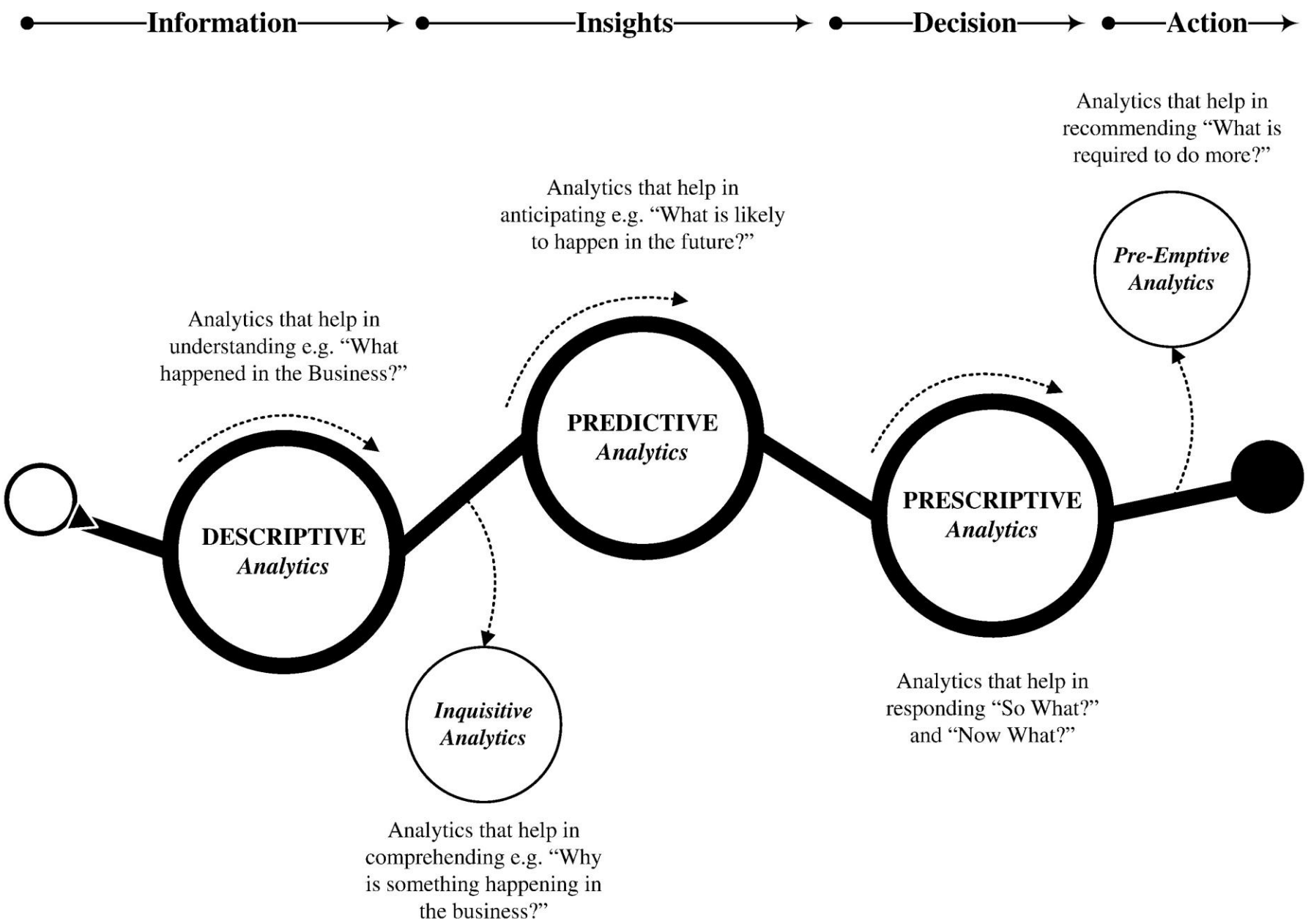
"Cerca de 60% dos nossos dados são provenientes de fontes estruturadas — bancos de dados e planilhas, por exemplo — e os 40% restantes são gerados a partir de fontes não estruturadas, como Twitter, Facebook e e-mail"
-- Roberto Zaina, REDE-LAB

- ✓ US$ 18 milhões em investimentos
- ✓ US$ 8,9 bilhões de ativos ilícitos identificados

COMO “FAZER” BIG DATA?



SIVARAJAH, Uthayasankar et al. Critical analysis of Big Data challenges and analytical methods. **Journal of Business Research**, v. 70, p. 263-286, 2017.



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COMO VOCÊ “FAZ” BIG DATA?